# Worksheet 3

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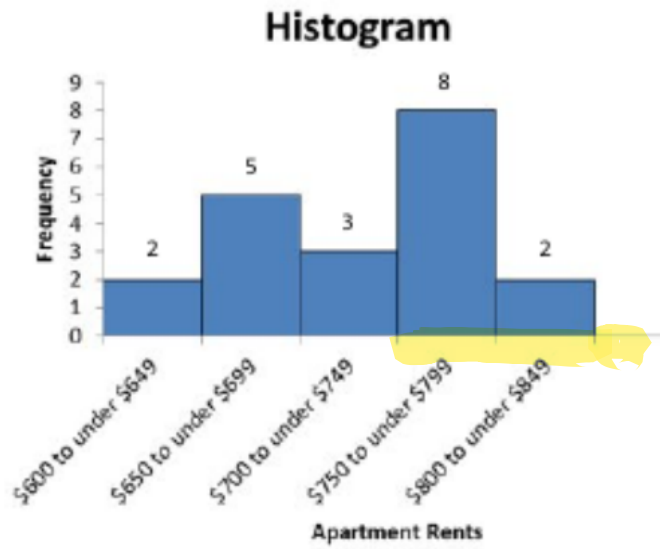
1. The following histogram shows the distribution of the monthly rental for a random sample of one-bedroom apartments in York, Pennsylvania.




freq 750 or above $= 8 + 2 = 10$

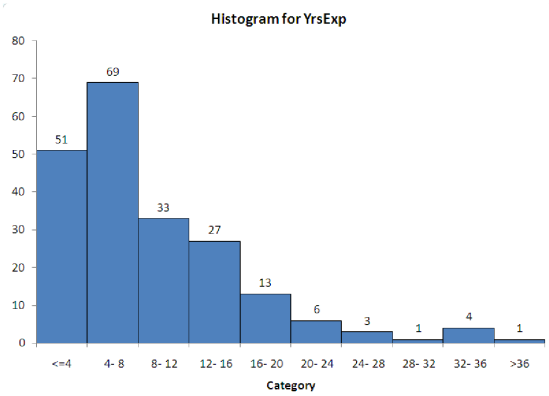
$2 + 5 + 3 + 8 + 2 = 20$

Rel freq 750 or above $= \frac{10}{20} = 0.5$

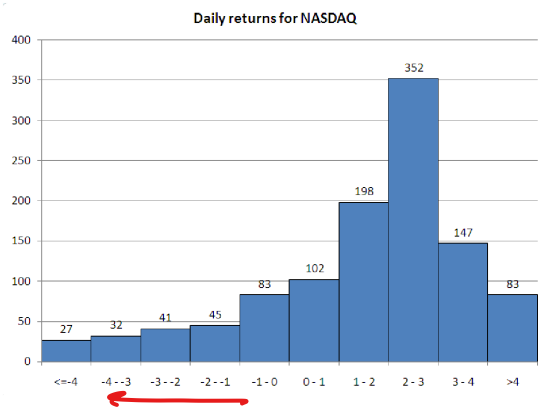
a. What is the total number of apartments in this sample, and what is the percentage of monthly rents that are $750 and above?

$0.5 \times 100 = 50\%$

b. Identify the shape of below histograms [1]




Postively skewed




negative skewed

[1] Graphs are from http://citadel.sjfc.edu/faculty/kgreen/MSTI130/MSTI130Text/Text_Fall_2014su28.html

2. The U.S. National Debt over the span of a decade from 1991 to 2001 is given in the following table:

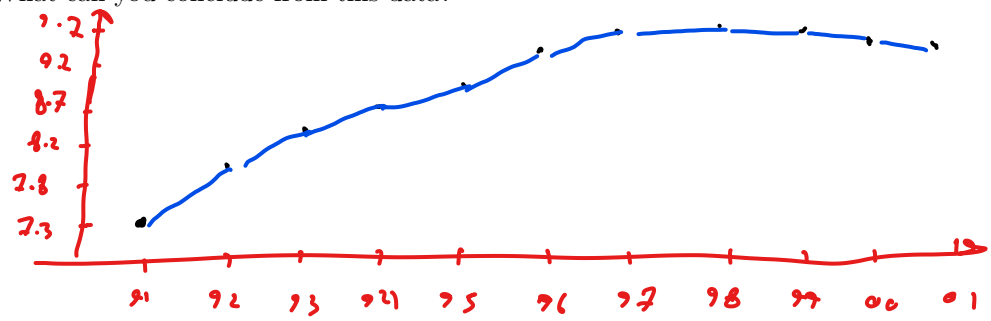
| Year | Debt (in T) |
|---|---|
| 1991 | 7.3 |
| 1992 | 7.9 |
| 1993 | 8.3 |
| 1994 | 8.6 |
| 1995 | 8.9 |
| 1996 | 9.1 |
| 1997 | 9.2 |
| 1998 | 9.3 |
| 1999 | 9.2 |
| 2000 | 9.0 |
| 2001 | 8.9 |

a. Is this an example of time series data or cross sectional data?

b. Make an appropriate plot for this data.

c. What can you conclude from this data?

b



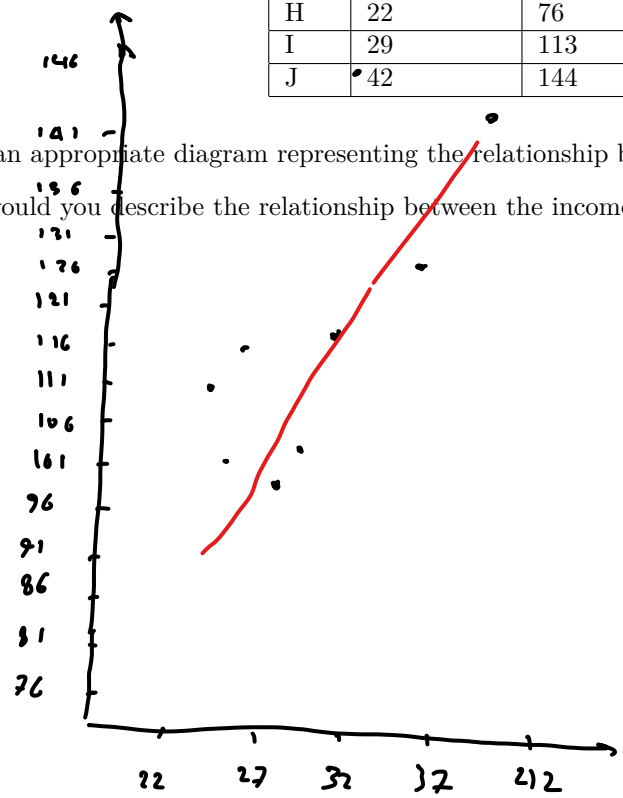
3. The following data has mean income and housing for 10 cities in Florida. Values are in dollars ($) and rounded to the nearest thousand.

| City | Income ( $x$) | Housing (y) |
|---|---|---|
| A | 26 | 109 |
| B | 29 | 97 |
| C | 25 | 115 |
| D | 28 | 99 |
| E | 38 | 122 |
| F | 32 | 145 |
| G | 25 | 100 |
| H | 22 | 76 |
| I | 29 | 113 |
| J | 42 | 144 |

a. Draw an appropriate diagram representing the relationship between Income (x) and Housing ($y$).

b. How would you describe the relationship between the income (x) and housing ($y$)? Positive

a)



4. A random sample of 10 students were asked how far they live from UMBC. These were their answers, rounded to the nearest half-mile:

$$1, 2.5, 1.5, 2, 0, 5, 3, 3.5, 1, 0.5$$

$n = 10$

a. Compute the mean, median and mode for this.

b. Compute the range.

(a) $\bar{x} = \frac{\sum x_i}{n} = \frac{1 + 2.5 + \cdots + 0.5}{10} = \frac{20}{10} = 2$

Median: 0 0.5 1 1 1.5 2 2.5 3 3.5 5

1 2 3 4 5 6

Pos Median $= (n+1) \div 0.5 = (10+1) \div 0.5 = 5.5$

Median $= \frac{1.5 + 2}{2} = 1.75$

Mode freq 1 : 2 $\Longrightarrow$ Mode = 1

freq any other = 1

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(b) Range = Max − Min = 5 − 0 = 5